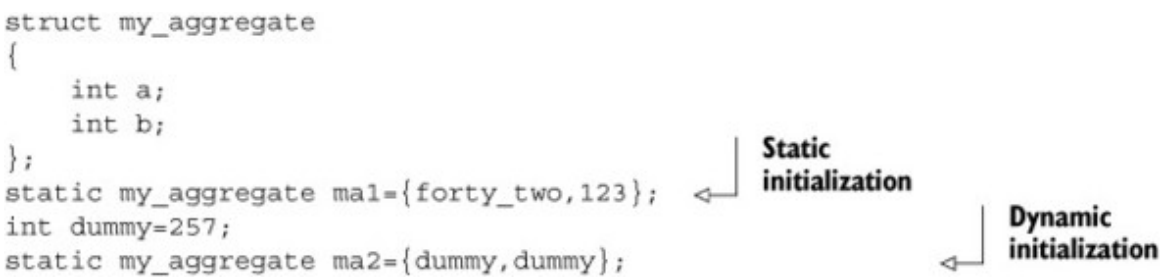


- Provide an initializer for a `static const` class data member of integral type in the class definition:

```
class X
{
static const int the_answer=forty_two;
};
```

- Provide an initializer for a built-in type or aggregate that can be used for static initialization:



- Static initialization like this can be used to avoid order-of-initialization problems and race conditions.

None of this is new—you could do all that with the 1998 edition of the C++ Standard. However, with the new Standard what constitutes a *constant expression* has been extended with the introduction of the `constexpr` keyword.

The `constexpr` keyword is primarily a function modifier. If the parameter and return type of a function meet certain requirements and the body is sufficiently simple, a function can be declared `constexpr`, in which case it can be used in constant expressions, for example:

```
constexpr int square(int x)
{
    return x*x;
}
int array[square(5)];
```